

Notes: Chernenko and Sunderam (RFS, 2014)
Frictions in Shadow Banking: Evidence from the Lending Behavior of Money
Market Mutual Funds

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1 Big picture

- **Question.** Can risk taking by shadow banks create negative spillovers to otherwise creditworthy borrowers? The paper studies whether money market mutual funds (MMMFs), which look like arm's-length investors, transmit distress across issuers through lending frictions.
- **Setting.** The empirical laboratory is the summer 2011 Eurozone sovereign debt episode. U.S. prime MMMFs had large exposures to Eurozone banks. When concerns about those banks intensified, especially after Moody's placed large French banks on review in June 2011, investors withdrew from exposed funds.
- **Core mechanism.** Funds with high Eurozone bank exposure suffered outflows. To meet redemptions, they reduced lending not only to Eurozone banks but also to non-European issuers. Other funds did not fully replace the lost funding because money-market lending relationships and preapproval frictions matter.
- **Central empirical contrast.** Holding the issuer fixed, a fund with greater Eurozone bank exposure cuts lending more than a fund with lower Eurozone bank exposure. This borrower fixed-effect design separates credit supply from issuer risk.
- **Main result.** Non-European issuers that relied on exposed funds raised less short-term financing from MMMFs after the shock. The effect is concentrated among less widely held issuers and among weak fund-issuer relationships.
- **Relationship result.** MMMF lending is not fully arm's length. Funds are less likely to cut strong relationships and only provide substitute financing to issuers with which they already have a relationship.
- **Economic takeaway.** Shadow-bank risk taking can create collateral damage. Even for large, highly rated firms in short-term credit markets, financing relationships and institutional frictions limit immediate substitution across lenders.

2 Incremental contribution of the paper

- **From bank lending channels to shadow-bank lending channels.** Prior evidence on credit supply frictions is strongest for traditional banks. This paper shows that a similar transmission channel operates inside market-based shadow banking.
- **From price spillovers to financing spillovers.** The institutional-investor spillover literature often focuses on temporary price pressure. This paper studies a real financing outcome: whether firms can raise short-term debt from MMMFs.
- **A complete chain of transmission.** The paper traces the sequence from incentives for MMMFs to take risk, to investor outflows, to reduced credit supply, to limited substitution across funds, and finally to issuer financing responses.
- **Novel security-level holdings data.** SEC Form N-MFP data allow the authors to observe fund-issuer-month positions. This is crucial because the identifying regressions compare lending by different funds to the same issuer.

- **Relationship evidence in an arm’s-length market.** The paper documents that relationships matter even in commercial paper and short-term debt markets, where borrowers are large and highly rated.
- **Policy contribution.** The results speak directly to MMMF regulation. Liquidity buffers, maturity restrictions, stable NAV rules, and capital-buffer proposals should be evaluated not only for investor-run risk but also for spillovers to borrowers.

3 Background and hypotheses

3.1 The 2011 Eurozone bank shock

Concerns about European sovereign debt began with Greece and then spread to other peripheral Eurozone countries. These sovereign concerns became bank concerns because Eurozone banks held sovereign debt and were indirectly exposed to weak Eurozone economies. The key event for the paper is June 15, 2011, when Moody’s placed BNP Paribas, Credit Agricole, and Societe Generale on review for downgrade because of their Greek exposures.

Interpretation: The paper uses this episode as a shock to the perceived riskiness of a specific subset of MMMF assets. The key point is not simply that the money market became risky; it is that funds differed in how much Eurozone bank exposure they had before the shock.

3.2 Pre-period and post-period

The paper separates the analysis into two symmetric three-month windows:

- **Pre-period:** March-May 2011. This is the period before the large investor redemptions.
- **Post-period:** June-August 2011. This is the period of large outflows from prime MMMFs after Eurozone bank concerns intensified.

Interpretation: This two-window design makes the empirical tests transparent. It also avoids relying too heavily on short-panel standard-error assumptions.

3.3 Main hypotheses

- **Hypothesis 1: risk-taking incentive.** Funds with more Eurozone bank exposure attract more inflows in the pre-period because Eurozone bank securities offer higher yields.
- **Hypothesis 2: run response.** Funds with more Eurozone bank exposure suffer larger outflows in the post-period.
- **Hypothesis 3: issuer-level spillover.** Issuers that rely on more Eurozone-exposed funds borrow less from MMMFs in the post-period.
- **Hypothesis 4: issuer-fixed-effect spillover.** Holding the issuer fixed, funds with greater Eurozone exposure cut lending more.
- **Hypothesis 5: relationship protection.** Funds under outflow pressure cut weak relationships more sharply than strong relationships.

- **Hypothesis 6: limited substitution.** Low-exposure funds provide replacement financing only to issuers with which they already have relationships.

Interpretation: The first two hypotheses establish the shock. The next two show the spillover. The last two identify the mechanism: lending relationships and institutional frictions.

4 Data and measurement

4.1 N-MFP holdings data

- The paper constructs security-level holdings of U.S. money market mutual funds from SEC Form N-MFP.
- The full dataset covers November 2010-May 2012, but most tests focus on March-August 2011.
- The main sample contains 177 prime MMMFs with about \$1.7 trillion in assets.
- The analysis focuses on prime funds because they can invest in nongovernment securities.
- Feeder funds, internal cash-management funds, variable annuities, prime muni funds, and funds entering or exiting the sample around the event are excluded.

Interpretation: The data contribution is essential. Without security-level positions, the paper could not compare how different funds treat the same issuer.

4.2 Issuer sample

- The paper collapses holdings to the fund-issuer-month level.
- Issuers are classified as ABCP, financial, government/agencies, municipal, nonfinancial, and other.
- For spillover tests, the authors exclude government, agency, supranational, municipal, and European issuers.
- The main non-European issuer sample contains 231 issuers: 72 ABCP, 75 financial, and 84 nonfinancial issuers.

Interpretation: The exclusion of European issuers is important. The paper wants to study indirect spillovers to non-European borrowers, not direct exposure to the European crisis.

4.3 Fund Euro share

The fund-level exposure measure is

$$Fund\ Euro\ share_{f,t} = \frac{\sum_{i \in Euro\ banks} Outstanding_{i,f,t}}{\sum_i Outstanding_{i,f,t}}.$$

This is the share of fund f 's portfolio invested in Eurozone banks at time t . In cross-sectional regressions, the authors use the pre-period average.

Interpretation: This variable measures how exposed each MMMF is to the shock. It is the key treatment-intensity variable at the fund level.

4.4 Issuer Euro share

The issuer-level exposure measure is

$$Issuer\ Euro\ share_{i,t} = \frac{\sum_f Outstanding_{i,f,t} \times Fund\ Euro\ share_{f,t}}{\sum_f Outstanding_{i,f,t}}.$$

This is the value-weighted average Eurozone exposure of the funds that lend to issuer i .

Interpretation: This variable measures how indirectly exposed an issuer is to the Eurozone bank shock through the identity of its money-market lenders.

4.5 Relationship measures

The paper adapts relationship-banking measures to the money-market setting:

- **In-sample frequency:** how often a fund lent to an issuer from November 2010-May 2011.
- **Out-of-sample relationship:** whether the fund held the issuer in earlier 2010 filings.
- **Maturity:** whether the fund's lending maturity to the issuer is longer than the issuer's median maturity.
- **Issuer-based quantity:** whether the issuer's portfolio share in the fund is above the median across funds for that issuer.
- **Fund-based quantity:** whether the issuer's portfolio share is above the median within that fund.

Interpretation: The relationship measures are not perfect, but they should be positively correlated with the strength of the lender-borrower relationship. Measurement error would bias against finding differences between strong and weak relationships.

5 Models and empirical specifications

5.1 Fund-flow regressions

The fund-flow regressions collapse the monthly panel into one fund-level cross-section. A representative specification is

$$Flow_f = \alpha + \beta Fund\ Euro\ share_f + \gamma Size_f + \varepsilon_f.$$

The dependent variable is cumulative fund flows scaled by beginning-of-period assets. The model is estimated separately for March-May 2011 and June-August 2011.

Interpretation: In the pre-period, a positive β means funds were rewarded for holding higher-yielding Eurozone bank exposure. In the post-period, a negative β means investors ran from exposed funds.

5.2 Issuer-level spillover regressions

At the issuer level, the paper estimates

$$\Delta Outstanding_i = \alpha + \beta Issuer\ Euro\ share_i + \varepsilon_i.$$

The dependent variable is the percentage change in the issuer’s average MMMF financing from the pre-period to the post-period.

Interpretation: A negative β means issuers funded by exposed money funds raised less financing after the shock. This is suggestive but not fully identified because issuers with high *Issuer Euro share* may differ from other issuers.

5.3 Fund-issuer fixed-effect regressions

The core identification specification is

$$\Delta Outstanding_{i,f} = \alpha_i + \beta Fund\ Euro\ share_f + \varepsilon_{i,f}.$$

The issuer fixed effect α_i compares different funds lending to the same issuer. The paper also controls for the issuer’s pre-period share in the fund’s portfolio.

Interpretation: This is the central credit-supply test. If Fund A and Fund B both lend to the same issuer, but Fund A has more Eurozone bank exposure, the design asks whether Fund A cuts lending more.

5.4 Exit regressions

The paper also studies an exit indicator:

$$Exit_{i,f} = 1\{\Delta Outstanding_{i,f} = -100\%\}.$$

Interpretation: This captures the extensive margin: whether the fund completely stops lending to the issuer.

5.5 Relationship regressions

The relationship tests interact fund Euro exposure with strong and weak relationship indicators:

$$\Delta Outstanding_{i,f} = \alpha_i + \beta_S(Fund\ Euro\ share_f \times Strong_{i,f}) + \beta_W(Fund\ Euro\ share_f \times Weak_{i,f}) + \dots$$

Interpretation: If $|\beta_W| > |\beta_S|$, exposed funds cut weak relationships more than strong relationships. This directly supports the relationship-lending mechanism.

5.6 Substitution across funds

The substitution test uses a balanced cross-section of all fund-issuer pairs:

$$\begin{aligned} \Delta PS_{i,f} = & \alpha_f + \beta_1 PS_{i,f,pre} + \beta_2 Existing_{i,f} \\ & + \beta_3(IssuerEuro_i \times Existing_{i,f}) \\ & + \beta_4(IssuerEuro_i \times NoExisting_{i,f}) + \varepsilon_{i,f}. \end{aligned}$$

Interpretation: Fund fixed effects compare issuers within the same fund. The test asks whether low-exposure funds increase lending to distressed issuers, and whether they do so only when they already know the issuer.

6 Identification and empirical design

6.1 What the design identifies

- The paper identifies credit-supply effects from MMMFs to issuers during the 2011 Eurozone-bank stress episode.
- The cleanest evidence comes from fund-issuer regressions with issuer fixed effects.
- This design rules out explanations based on time-varying issuer risk that affects all funds lending to the same issuer in the same way.

Interpretation: The core empirical question is not whether risky issuers lost funding. It is whether the same issuer lost more funding from exposed funds than from less exposed funds.

6.2 Why issuer fixed effects matter

Issuer-level regressions can be confounded by issuer quality. For example, risky non-European issuers might both borrow from risky MMMFs and lose funding in 2011. The issuer-fixed-effect design removes this concern by comparing lenders to the same borrower.

Interpretation: This mirrors the logic of bank-lending-channel papers such as Khwaja and Mian. The paper imports that design into the shadow-banking setting.

6.3 Mechanism evidence

The paper does not stop at documenting that exposed funds cut lending. It shows:

- The effect is stronger for less widely held issuers, where frictions are more plausible.
- Exposed funds cut weak relationships more than strong relationships.
- Low-exposure funds substitute only for issuers with existing relationships.

Interpretation: These mechanism tests separate the paper from a generic run story. The key mechanism is not merely that funds lose assets; it is that replacement credit is relationship-constrained.

6.4 Limitations of the design

- The paper observes MMMF financing, not every possible source of short-term funding.
- The event is a moderate stress episode, not a full-scale crisis like September 2008.
- The issuer-level real-outcome tests have small samples, especially for nonfinancial firms with consistent commercial-paper disclosure.

- The results identify a channel, but the magnitudes could be larger or smaller in a more severe stress episode.

Interpretation: The safest interpretation is that the paper documents a transmission mechanism. It does not claim that the 2011 episode produced large real investment effects for all affected firms.

7 Tables and figures

7.1 Figure 1: Total net assets of prime money market mutual funds

Read:

- The figure plots weekly total net assets for institutional and retail prime MMMFs from March 2011 to July 2012.
- The vertical line marks Moody's review of large French banks in June 2011.
- Institutional prime MMMF assets drop sharply after the review, while retail assets are much flatter.
- The fall in aggregate prime MMMF assets was about \$180 billion, or roughly 11%, between early June and the end of August 2011.

Economic message: The figure gives the visual shock. The run was not a one-day panic like September 2008, but a slow and economically large institutional outflow from prime MMMFs.

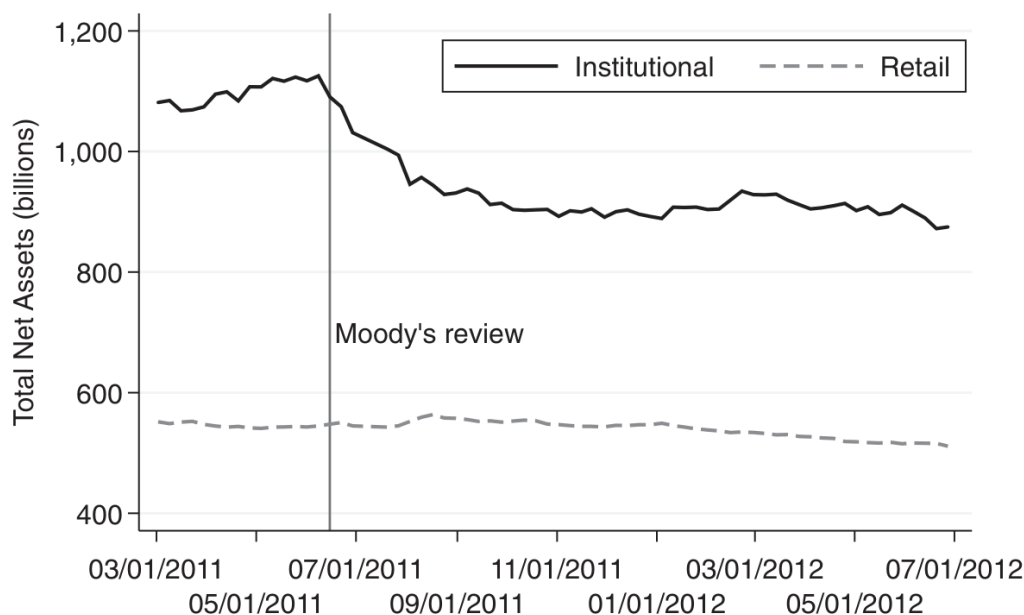


Figure 1
Total net assets of prime money market mutual funds
 Weekly data from the Investment Company Institute.

Figure 1: total net assets of prime money market mutual funds.

7.2 Table 1: Summary statistics for funds

Read:

- The average fund has about \$8.95 billion in total net assets, but fund size is highly skewed.
- The average portfolio maturity is about 39 days, consistent with the short maturity of MMMF assets.
- Eurozone financial issuers represent 17.9% of the average fund portfolio, and the general Euro share measure averages about 17.0%.
- Financial issuers dominate the portfolio: ABCP, certificates of deposit, financial commercial paper, and non-EU financial issuers account for large shares.
- Net yields are extremely low, with a mean around 3.8 basis points, which makes small yield differences valuable to investors searching for yield.

Economic message: The average prime MMMF is highly exposed to financial issuers and materially exposed to Eurozone banks. This makes the sector a plausible channel through which Eurozone bank distress can affect U.S. short-term credit markets.

Table 1
Summary statistics: Funds

			Percentile		
	Mean	SD	25	50	75
Fund characteristics					
Total net assets (millions)	8,950	18,571	535	1,869	8,957
Institutional share (%)	54.8	45.0	0.0	68.6	100.0
Portfolio maturity (days)	38.7	11.1	33.0	41.0	46.0
Gross yield (bps)	23.7	7.5	18.0	24.0	29.0
Net yield (bps)	3.8	5.6	1.0	1.0	5.0
Euro share (%)	17.0	11.3	7.7	16.9	24.4
Unsecured Euro share (%)	12.6	9.4	4.8	11.2	19.0
Fund flows (bps)	−43.2	748.7	−369.3	−36.5	266.6
Instrument shares (%)					
ABCP	11.1	11.9	0.8	7.5	18.4
CD	21.6	16.3	6.4	20.5	34.4
Financial CP	16.1	9.4	10.5	14.6	20.4
Government/agency	12.5	12.3	3.6	10.2	18.1
Government/agency repo	11.9	13.0	1.7	8.6	16.6
Nonfinancial CP	8.2	13.4	0.0	2.3	10.1
Municipal debt	5.9	9.1	0.0	1.8	9.2
Other repo	2.8	5.8	0.0	0.0	2.2
Other	9.8	11.1	3.4	7.4	12.4
Issuer shares (%)					
ABCP	10.5	11.6	0.0	7.2	17.5
Government/agency	12.8	12.6	3.8	10.5	18.1
Rest of EU financial	11.4	7.6	5.8	10.8	16.1
Eurozone financial	17.9	11.8	8.2	17.6	25.1
Municipal	6.3	10.2	0.0	1.9	9.2
Non-EU financial	28.2	13.0	19.1	28.8	37.6
Nonfinancial	9.5	14.7	0.7	3.6	10.7
Other	3.3	8.6	0.0	1.6	3.9

Table 1: summary statistics for funds.

7.3 Table 2: Summary statistics for issuers and fund-issuer exposures

Read:

- The sample includes 231 non-European private issuers observed across issuer-months and fund-issuer-months.
- The median issuer has much less outstanding than the mean issuer, indicating a skewed issuer-size distribution.
- The median issuer is held by nine funds, while some issuers are held by many funds.
- The average fund-issuer exposure is about \$150 million, and the median exposure is \$35 million.
- The average fund-issuer portfolio share is about 1.9%, leaving room for portfolio concentration constraints to matter.

Economic message: Table 2 shows why relationship frictions are plausible. Some issuers are broadly held and close to frictionless, but many issuers depend on a relatively small set of funds.

Table 2
Summary statistics: Issuers and exposures

	<i>N</i>	Mean	SD	Percentile		
				25	50	75
Panel A: Issuers						
Outstanding (millions)	1,386	2980.4	7707.8	34.8	327.3	1632.2
Weighted average maturity (days)	1,264	49.4	72.5	12.5	26.2	46.2
Yield (bps)	1,251	25.5	11.8	17.5	25.0	32.4
Issuer Euro share (%)	1,264	16.0	7.9	9.9	16.6	21.4
Number of funds	1,386	21.9	28.3	4.0	9.0	30.0
Panel B: Fund-issuer exposures						
Exposure (millions)	24,905	150.4	398.0	10.0	35.0	126.9
Portfolio share (%)	24,905	1.9	1.9	0.6	1.3	2.6
Weighted average maturity (days)	24,904	42.1	55.3	9.0	25.3	53.6
Yield (bps)	24,283	24.6	11.6	17.1	24.2	30.4

This table reports summary statistics for issuers and fund-issuer exposures in our data. The sample of issuers consists of non-European issuers, excluding sovereign, agency, municipal, and supranational issuers. The sample of funds consists of prime money market mutual funds filing form N-MFP during the whole March–August 2011 period, excluding funds that exclusively serve the internal cash management needs of their mutual fund families, feeder funds that invest in other funds, variable annuities, and prime muni funds.

Table 2: summary statistics for issuers and fund-issuer exposures.

7.4 Table 3: Fund flows during the pre- and post-periods

Read:

- Columns 1-5 study March-May 2011. Higher net yield predicts higher inflows, consistent with a performance-flow relation.
- In the pre-period, Fund Euro share is positively related to flows, consistent with investors rewarding higher-yielding Eurozone bank exposure before the shock.

- Columns 6-10 study June-August 2011. Fund Euro share becomes negative and significant: exposed funds suffer larger outflows.
- The effect is driven by unsecured Eurozone exposure, not repo exposure.
- The outflow sensitivity is especially strong for institutional funds.

Economic message: Table 3 establishes both sides of the risk-taking mechanism. Eurozone bank exposure was attractive before the shock because it raised yield, but it became a liability once investors focused on Eurozone bank risk.

Table 3
Fund flows during the pre- and post-periods

	March–May 2011					June–August 2011				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Size	2.88*	3.09*	2.02	3.00*	2.60	−8.13***	−6.65***	−6.50***	−5.88***	−6.08***
	(1.67)	(1.58)	(1.71)	(1.61)	(1.64)	(2.07)	(2.06)	(2.14)	(2.04)	(2.16)
Net yield (bps)	1.38***		1.27**		−0.82	−0.40		−0.19		−2.50*
	(0.52)		(0.53)		(0.69)	(0.98)		(0.96)		(1.47)
Fund Euro share		0.52*	0.42		0.51		−0.88***	−0.87***		−0.15
		(0.27)	(0.28)		(0.31)		(0.31)	(0.32)		(0.38)
Unsecured Euro share				0.56*					−1.29***	
				(0.30)					(0.38)	
Repo Euro share				0.46					−0.03	
				(0.50)					(0.54)	
Institutional fund					9.96					37.01**
					(11.77)					(17.51)
Fund Euro share					−0.53					−2.08***
× Institutional					(0.61)					(0.65)
Net yield					2.62***					3.01*
× Institutional					(0.93)					(1.81)
Constant	−32.05***	−36.56***	−33.17***	−36.15***	−36.32***	57.98***	61.79***	61.22***	57.49***	50.34***
	(11.40)	(11.13)	(11.49)	(11.53)	(11.14)	(15.84)	(15.16)	(15.40)	(15.00)	(15.19)
<i>N</i>	177	177	177	177	177	177	177	177	177	177
Adjusted <i>R</i> ²	0.082	0.057	0.094	0.052	0.135	0.093	0.124	0.120	0.135	0.160

In Columns 1–5, the dependent variable is cumulative net flows during the March–May 2011 period scaled by February 2011 net assets. In Columns 6–10, the dependent variable is cumulative net flows during the June–August 2011 period scaled by May 2011 net assets. Fund flows are annualized and winsorized at the 5th and 95th percentiles. Explanatory variables are measured as of the beginning of the relevant period, i.e., as of February 2011 in Columns 1–5 and as of May 2011 in Columns 6–10. Fund flows and Euro share are expressed in percentage form. Fund Euro share is the share of the fund's portfolio invested in Eurozone banks. Unsecured Euro share is the fraction of the fund's portfolio invested in unsecured Eurozone bank claims. Repo Euro share is the fraction of repurchase agreements with Eurozone banks in the fund's portfolio. Fund size is the log of fund TNA. Institutional funds are funds with more than 99% of fund assets in institutional share classes. Robust standard errors are reported. *, **, and *** indicate statistical significance at 10%, 5%, and 1%.

Table 3: fund flows during the pre- and post-periods.

7.5 Table 4: Issuer-level effects

Read:

- The dependent variable is the percentage change in issuer-level MMMF financing from the pre-period to the post-period.
- Issuer Euro share is negative and statistically significant across all specifications.
- A 10 percentage point higher Issuer Euro share is associated with roughly 14-16 percentage points less growth in MMMF financing.
- Adding issuer-type fixed effects and controlling for issuer yield does not remove the result.

Economic message: Issuers whose money-market lenders were more exposed to Eurozone banks raised less financing after the shock. This is the first direct spillover result, although it still may mix credit supply with issuer-level risk.

7.6 Table 6: Persistence of spillover effects

Read:

- Table 6 repeats the fund-issuer analysis using later post-period windows.
- The Fund Euro share coefficient is strongly negative through August 2011 and weaker by November 2011.
- By February and May 2012, the coefficient is near zero and statistically insignificant.
- The effect therefore fades gradually rather than reversing immediately.

Economic message: The spillover is short-run but not instantaneous. The gradual decay is consistent with lending frictions that take time to overcome.

Table 4
Issuer-level effects

	(1)	(2)	(3)
Issuer Euro share _{<i>i</i>} (%)	-1.611*** (0.585)	-1.473** (0.619)	-1.407** (0.642)
Yield _{<i>i</i>} (%)			-0.157 (0.473)
Constant	0.344*** (0.128)	0.319** (0.129)	0.348** (0.163)
Issuer type FE	-	+	+
<i>N</i>	231	231	231
Adjusted <i>R</i> ²	0.037	0.030	0.027

(a) Table 4: issuer-level effects.

Table 6
Persistence of spillover effects

	Change from pre-period to 3 months ending			
	August 2011	November 2011	February 2012	May 2012
	(1)	(2)	(3)	(4)
Portfolio share _{<i>i, f</i>} (%)	-14.42*** (1.44)	-17.35*** (1.78)	-20.11*** (2.08)	-19.06*** (1.96)
Fund Euro share _{<i>f</i>} (%)	-0.86*** (0.28)	-0.64* (0.34)	-0.13 (0.38)	-0.06 (0.36)
Constant	36.17*** (5.65)	31.08*** (7.60)	21.86*** (8.22)	10.34 (7.40)
<i>N</i>	3837	3837	3837	3837
Adjusted <i>R</i> ²	0.116	0.154	0.182	0.191

(b) Table 6: persistence of spillover effects.

7.7 Table 5: Identifying spillovers to non-European issuers

Read:

- Table 5 is the core identification table. The regressions include issuer fixed effects, so they compare lending by different funds to the same issuer.
- For the top ten most widely held issuers, Fund Euro share has no meaningful effect.
- For other issuers, higher Fund Euro share predicts a larger reduction in lending.
- In the baseline specification for other issuers, a 10 percentage point higher Fund Euro share implies roughly 8.6 percentage points more reduction in exposure.
- Panel B shows the extensive margin: high-Euro-share funds are more likely to completely exit a position.
- Unsecured Eurozone exposure drives the result; repo exposure does not.

Economic message: Table 5 provides the cleanest evidence of credit-supply spillovers. Holding the borrower fixed, exposed funds reduce lending more. The fact that the result is absent for the most widely held issuers supports the interpretation that frictions matter most where substitution is harder.

Table 5
Identifying spillovers to non-European issuers

	Top 10	Other issuers			
	(1)	(2)	(3)	(4)	(5)
Panel A: $\Delta Outstanding_{i,f}$ (%)					
Portfolio share $_{i,f}$ (%)	-13.59*** (1.83)	-14.42*** (1.44)	-14.40*** (1.45)	-14.40*** (1.45)	-14.56*** (1.45)
Fund Euro share $_f$ (%)	-0.09 (0.26)	-0.86*** (0.28)	-0.87*** (0.28)	-0.74 (0.62)	
Yield $_{i,f}$ (bps)			0.08 (0.26)	0.17 (0.46)	
Fund Euro share $_f \times$ Yield $_{i,f}$				-0.01 (0.02)	
Unsecured Euro share $_f$ (%)					-1.11*** (0.31)
Repo Euro share $_f$ (%)					-0.18 (0.49)
Constant	50.98*** (6.86)	36.17*** (5.65)	34.23*** (9.14)	32.25** (13.25)	36.97*** (5.64)
N	1133	3837	3837	3837	3837
Adjusted R^2	0.103	0.116	0.115	0.115	0.117
Panel B: $Exit_{i,f}$ (%)					
Portfolio share $_{i,f}$ (%)	-2.07*** (0.43)	-3.39*** (0.53)	-3.56*** (0.53)	-3.57*** (0.53)	-3.35*** (0.53)
Fund Euro share $_f$ (%)	0.19 (0.12)	0.38*** (0.12)	0.39*** (0.12)	0.55** (0.25)	
Yield $_{i,f}$ (bps)			-0.48*** (0.07)	-0.37** (0.14)	
Fund Euro share $_f \times$ Yield $_{i,f}$				-0.01 (0.01)	
Unsecured Euro share $_f$ (%)					0.47*** (0.12)
Repo Euro share $_f$ (%)					0.14 (0.25)
Constant	6.94** (2.71)	16.81*** (2.63)	28.75*** (3.54)	26.25*** (4.97)	16.53*** (2.60)
N	1133	3837	3837	3837	3837
Adjusted R^2	0.040	0.156	0.166	0.166	0.156

Table 5: identifying spillovers to non-European issuers.

7.8 Table 7: Relationships in money markets

Read:

- Table 7 interacts Fund Euro share with strong and weak relationship indicators.
- Across relationship measures, exposed funds cut weak relationships more strongly than strong relationships.
- In the quantity-based measures, the effect of Fund Euro share on lending reductions is roughly

twice as large for weak relationships as for strong relationships.

- Panel B shows the same logic for exits: exposed funds are more likely to exit weak relationships.
- Strong relationships are also directly associated with lower exit probabilities.

Economic message: The table shows that money-market finance has relationship-lending features. When a fund is constrained by outflows, it protects stronger borrower relationships and sacrifices weaker ones.

Table 7
Relationships in money markets

	Length/frequency		Maturity	Quantity	
	In sample (1)	Out of sample (2)	(3)	Issuer based (4)	Fund based (5)
Panel A: $\Delta Outstanding_{i,f}$					
Portfolio share $_{i,f}$ (%)	-14.01*** (1.43)	-14.37*** (1.44)	-15.03*** (1.50)	-7.78*** (1.26)	-9.19*** (1.18)
Strong relationship $_{i,f}$	-7.27 (8.70)	-4.99 (7.85)	21.91*** (7.51)	-45.10*** (8.86)	-46.35*** (8.21)
Fund Euro share $_{i,f}$	-0.83*** (0.28)	-0.74** (0.32)	-0.69** (0.30)	-0.74*** (0.25)	-0.52** (0.23)
× Strong relationship $_{i,f}$	-0.97** (0.41)	-0.96*** (0.35)	-0.96*** (0.36)	-1.54*** (0.44)	-1.20*** (0.44)
N	3837	3837	3837	3837	3837
Adjusted R^2	0.116	0.115	0.134	0.133	0.136
Panel B: $Exit_{i,f}$					
Portfolio share $_{i,f}$ (%)	-1.80*** (0.44)	-3.18*** (0.52)	-3.01*** (0.52)	-1.57*** (0.44)	-0.96** (0.48)
Strong relationship $_{i,f}$	-16.76*** (3.29)	-6.18* (3.23)	-15.06*** (3.04)	-5.56 (3.59)	-11.46*** (3.32)
Fund Euro share $_{i,f}$	0.21* (0.11)	0.23* (0.12)	0.32** (0.14)	0.27* (0.14)	0.28** (0.11)
× Strong relationship $_{i,f}$	0.41** (0.16)	0.45*** (0.16)	0.40*** (0.15)	0.42*** (0.16)	0.53*** (0.17)
N	3837	3837	3837	3837	3837
Adjusted R^2	0.214	0.169	0.200	0.163	0.184

Table 7: relationships in money markets.

7.9 Table 8: Substitution across funds

Read:

- Table 8 asks whether less-exposed funds provide replacement financing to affected issuers.
- The dependent variable is the change in the issuer's portfolio share within a fund.
- For low-Euro-share funds, the interaction of Issuer Euro share with existing relationship is positive and significant.

- The interaction with no existing relationship is not positive in the low-exposure-fund specification.
- High-exposure funds do not provide the same offsetting substitution.

Economic message: Substitute financing exists, but only through existing relationships. This is direct evidence that affected issuers cannot simply tap any unconstrained MMMF at once.

7.10 Table 9: Long-term bond issuance, cash, commercial paper, and capital expenditures

Read:

- Table 9 studies nonfinancial issuers for which the authors can observe additional financing and investment variables.
- Issuers with higher exposure to affected MMMFs issue more long-term debt in Q2 and Q3.
- Their total commercial paper outstanding falls in Q2 and Q3.
- Changes in cash are generally negative but statistically insignificant.
- Capital expenditures do not show a statistically significant response.

Economic message: The evidence suggests that affected nonfinancial firms partially substitute away from commercial paper toward long-term debt. The paper finds a financing adjustment, but not clear investment effects in this moderate stress episode.

Table 8
Substitution across funds

	Fund Euro share		
	All funds (1)	Low (2)	High (3)
Portfolio share _{<i>i, f</i>} (bps)	-0.20*** (0.02)	-0.12* (0.06)	-0.29*** (0.06)
Existing relationship _{<i>i, f</i>}	3.37 (6.90)	-14.45 (12.71)	7.90 (9.80)
Issuer Euro share _{<i>i</i>} × Existing relationship _{<i>i, f</i>} (%)	0.52 (0.40)	1.23** (0.51)	0.44 (0.58)
Issuer Euro share _{<i>i</i>} × No existing relationship _{<i>i, f</i>} (%)	-0.02 (0.03)	-0.07** (0.03)	0.05* (0.03)
Constant	2.22*** (0.50)	3.52*** (0.77)	0.64* (0.38)
<i>N</i>	40710	20470	20240
Adjusted <i>R</i> ²	0.082	0.031	0.166

(a) Table 8: substitution across funds.

Table 9
Long-term bond issuance, changes in cash and capital expenditures

	(1)	(2)	(3)	(4)	(5)	(6)
	Long-term debt issuance			Change in cash		
	Q2	Q3	Q4	Q2	Q3	Q4
Issuer Euro share _{<i>i</i>}	0.14** (0.07)	0.13* (0.07)	0.07 (0.10)	0.03 (0.08)	-0.07 (0.10)	-0.08 (0.09)
Constant	-0.50 (0.42)	-0.51 (0.42)	-0.03 (0.64)	0.05 (0.54)	-0.11 (0.68)	0.51 (0.66)
<i>N</i>	42	42	42	42	42	42
<i>R</i> ²	0.039	0.035	0.004	0.001	0.004	0.005
	Change in CP			Change in CapEx		
	Q2	Q3	Q4	Q2	Q3	Q4
Issuer Euro share _{<i>i</i>}	-0.10** (0.05)	-0.14* (0.07)	-0.04 (0.09)	-0.06 (0.04)	-0.01 (0.02)	0.00 (0.02)
Constant	0.02 (0.29)	0.60 (0.38)	-0.27 (0.53)	0.51* (0.26)	0.22** (0.09)	0.30*** (0.10)
<i>N</i>	42	42	42	42	42	42
<i>R</i> ²	0.034	0.039	0.002	0.022	0.008	0.000

(b) Table 9: long-term debt, cash, CP, and CapEx.

7.11 Appendix Table A1: Variable definitions

Read:

- Table A1 defines the main fund-level, issuer-level, and fund-issuer-level variables.
- It clarifies the construction of Fund Euro share, Issuer Euro share, flows, portfolio share, maturity, relationship strength, and yield.
- The appendix is especially useful because the empirical design uses several levels of observation: fund, issuer, and fund-issuer pair.

Economic message: The variable definitions make clear that the paper is a matched borrower-lender analysis. The empirical contribution depends on measuring treatment exposure at the lender level and spillover exposure at the borrower level.

Table A1
Variable Definitions

<i>Fund Euro share</i> _{<i>f,t</i>}	The share of fund <i>f</i> 's portfolio invested in Eurozone banks in month <i>t</i> . In cross-sectional regressions, we calculate the average value during the pre-period, <i>Fund Euro share</i> _{<i>f</i>} .
<i>Fund flows</i> _{<i>f,t</i>}	Net subscriptions scaled by lagged assets, $\frac{Flows_{f,t}}{TNA_{f,t-1}}$. In cross-sectional regressions, we calculate cumulative fund flows during the pre- (post) period, scaled by February (May) 2011 assets. Fund flows are annualized and winsorized at the 5th and 95th percentiles.
<i>Gross yield</i> _{<i>f,t</i>}	The fund's 7-day gross yield reported on form N-MFP. Some funds incorrectly report their gross yield as zero, which cannot be the case. These observations are excluded from the regressions that include gross yield.
<i>Institutional fund</i> _{<i>f</i>}	Binary variable equal to one for funds with <i>Institutional share</i> greater than 99%.
<i>Institutional share</i> _{<i>f,t</i>}	The share of fund's assets in institutional share classes. A share class is considered to be institutional if (1) its minimum initial investment is equal to or greater than \$1 million or is equal to \$1, or (2) the name of the share class includes "institutional."
<i>Issuer Euro share</i> _{<i>i,t</i>}	The value-weighted average of <i>Fund Euro shares</i> , calculated over all funds holding issuer <i>i</i> at time <i>t</i> , with the fraction of issuer <i>i</i> held by fund <i>f</i> as the weight. $Issuer\ Euro\ share_{i,f,t} = \sum_f \frac{Outstanding_{i,f,t}}{\sum_f Outstanding_{i,f,t}} \times Fund\ Euro\ share_{f,t}$. In cross-sectional regressions, we calculate the average value during the pre-period, <i>Issuer Euro share</i> _{<i>i</i>} .
<i>Net yield</i> _{<i>f,t</i>}	The value-weighted average of the 7-day net yields on the fund's share classes.
$\Delta Outstanding_i$	Percentage change in the average outstanding of issuer <i>i</i> between the pre- and post-periods. Variable is winsorized at the 5th and 95th percentiles.
$\Delta Outstanding_{i,f}$	Percentage change in the average exposure of fund <i>f</i> to issuer <i>i</i> between the pre- and post-periods. Variable is winsorized at the 5th and 95th percentiles.
<i>Portfolio maturity</i> _{<i>f,t</i>}	The fund's dollar-weighted average portfolio maturity.
<i>Relationship strength (in-sample)</i> _{<i>i,f</i>}	The number of months between November 2010 and May 2011 in which fund <i>f</i> has a position in issuer <i>i</i> , divided by the number of months in which issuer <i>i</i> is held by any prime money market fund. A fund is considered to have a strong relationship with an issuer if it lends more consistently (frequently) than the median fund lending to that issuer.
<i>Relationship strength (out-of-sample)</i> _{<i>i,f</i>}	Binary variable equal to one whenever fund <i>f</i> holds issuer <i>i</i> during the first six months of 2010. Fund holdings are from the last N-Q or N-CSR(S) filing made during this period.
<i>Relationship strength (maturity)</i> _{<i>i,f</i>}	A fund-issuer pair is considered to have a strong relationship if the maturity of the issuer's borrowing from the fund is longer than the median for that issuer.
<i>Relationship strength (issuer-based quantity)</i> _{<i>i,f</i>}	A fund-issuer pair is considered to have a strong relationship if the portfolio share is above the median for that issuer.
<i>Relationship strength (fund-based quantity)</i> _{<i>i,f</i>}	A fund-issuer pair is considered to have a strong relationship if the portfolio share is above the median for that fund.
<i>Repo Euro share</i> _{<i>f,t</i>}	The fraction of repurchase agreements with Eurozone banks in the fund's portfolio.
<i>Size</i> _{<i>f,t</i>}	The log of fund's TNA.
<i>Unsecured Euro share</i> _{<i>f,t</i>}	The difference between <i>Fund Euro share</i> _{<i>f,t</i>} and <i>Repo Euro share</i> _{<i>f,t</i>} . Measures the fraction of unsecured claims on Eurozone banks in the fund's portfolio.
<i>Yield</i> _{<i>i,t</i>}	The weighted-average yield paid by issuer <i>i</i> at time <i>t</i> . We extract the yield on each security from the title of the issue, reported in item 27 of form N-MFP. When the issue title does not include its yield, we calculate it based on the principal (item 40), amortized cost (item 41), and time to maturity (item 35).

Appendix Table A1: variable definitions.

8 Short summary

- The paper studies U.S. prime MMMFs during the 2011 Eurozone bank stress episode.
- Funds with higher Eurozone bank exposure earned higher yields and attracted more flows before the shock.
- After the shock, those same funds suffered larger outflows, especially institutional funds with unsecured Eurozone bank exposure.
- Non-European issuers funded by exposed MMMFs raised less short-term financing from MMMFs.
- The core identification compares lending by different funds to the same issuer; exposed funds cut lending more.
- The effects are stronger for less widely held issuers and weak fund-issuer relationships.
- Low-exposure funds provide substitute financing only to issuers with which they already have relationships.
- Nonfinancial issuers appear to substitute partly into long-term debt, while the paper finds no clear capital-expenditure effect in this moderate episode.
- The main lesson is that shadow-bank lending is not frictionless arm's-length finance.

9 Conclusion

9.1 Core conclusion

Chernenko and Sunderam show that risk taking by money market mutual funds can create spillovers to other borrowers. Funds that invested heavily in Eurozone banks experienced outflows in summer 2011. Those outflows reduced the credit supply available to non-European issuers that relied on the same funds. Because substitution across MMMFs was relationship-constrained, the affected issuers could not immediately replace all lost financing.

9.2 Economic intuition

The economic intuition is similar to a bank lending channel, but it operates through shadow banks. A lender takes risk by financing a troubled borrower. When the risky exposure becomes salient, the lender loses funding. To shrink its portfolio, the lender cuts credit to other borrowers. Those other borrowers are not necessarily risky, but they are affected because they depend on the same lender. In a frictionless capital market, other lenders would immediately replace the lost funding. In this paper, they do not, because MMMFs rely on prior relationships, internal approval lists, and limited short-run credit research.

9.3 Incremental contribution revisited

The paper advances the literature in three ways. First, it extends lending-channel logic from traditional banks to shadow banks. Second, it shows that relationship frictions matter in money markets, even for large and highly rated borrowers. Third, it connects investor runs on MMMFs to borrower financing outcomes, not only to asset prices or fund-level flows.

9.4 Implications for regulation

The evidence suggests that MMMF regulation should consider borrower-side spillovers. Rules that reduce run incentives may reduce forced portfolio contraction. Rules that make funds safer by shortening asset maturities may have ambiguous effects because issuers must refinance more frequently. Capital buffers, floating NAVs, and liquidity requirements all matter not only for fund investors but also for the stability of short-term corporate financing.

9.5 Limitations

The episode is moderate relative to the 2008 crisis, so the paper identifies a channel rather than the maximum possible effect. The main data cover financing from MMMFs, not all short-term credit sources. The real-outcome evidence is limited by the small nonfinancial sample and by voluntary commercial-paper disclosure. Relationship measures are proxies, not direct observations of internal fund approval processes.

9.6 Takeaway

The enduring takeaway is that arm's-length finance is not fully arm's length. In short-term debt markets, lender identity matters. Shadow banks can transmit distress because their risk taking affects their own funding stability, and because borrowers cannot instantly move across lenders when relationships and institutional frictions constrain substitution.